

Speech-to-Text, Translation & Text-to-Speech



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Project Overview & Goals



The Goal

Take spoken English and deliver it back as speech in another language.

Target languages:

Arabic (ar) **French** (fr) **Spanish** (es)

Tested across three conditions — short, long, and noisy audio.



Accurate transcription

Convert speech to text reliably, even with background noise.



Faithful translation

Preserve meaning, and grammar across languages.



Natural-sounding output

Produce intelligible speech with correct native pronunciation.

How the Pipeline Works



Input Audio

English speech clip
(short, long, or noisy)
loaded from Google
Drive.



Speech-to-Text

Whisper (small)
transcribes the audio
into English text.



Translation

Google Translate
converts the text to
Arabic, French, and
Spanish.



Text-to-Speech

gTTS synthesizes the
translated text into a
spoken MP3.



Tools: OpenAI Whisper · deep-translator (Google Translate) · gTTS · ffmpeg — run on a Colab T4 GPU

Modules & Tools Used



Speech-to-Text

openai-whisper

Transcribes English audio into text.
Loaded as the "small" model on GPU.



Translation

deep-translator

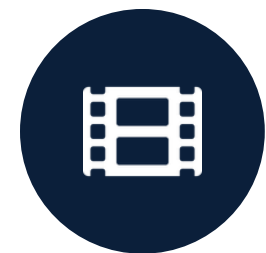
GoogleTranslator wrapper — converts
English text to Arabic, French & Spanish.



Text-to-Speech

gTTS

Google Text-to-Speech — synthesizes
translated text into spoken MP3 files.



Audio decoding

ffmpeg

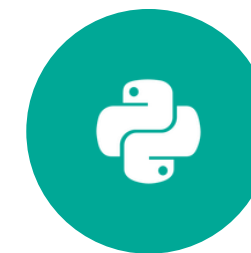
Backend that lets Whisper read the
.mp4 audio inputs.



Storage

google.colab.drive

Mounts Google Drive to load input clips
and save generated audio.



Playback & timing

IPython.display / time

Plays audio inline and measures per-
stage latency.

The ASR & TTS Algorithms



ASR — OpenAI Whisper

Transformer encoder–decoder

Trained on a large, multilingual corpus of audio.

How it processes sound:

- Audio → log-Mel spectrogram
- Encoder builds a representation of the sound
- Decoder generates text token-by-token
- Each word predicted from the audio and the words already decoded
→ *context-aware and robust to noise.*



TTS — gTTS (Google TTS)

Neural speech synthesis

Sends text to Google's Translate TTS engine.

How it produces speech:

- Text is sent to Google's TTS service
- A neural model renders natural-sounding speech
- Returned as a ready-to-play MP3 file
- Native pronunciation for Arabic, French & Spanish
→ *one engine covers all three target languages.*

Results — Speech Recognition

Whisper's English transcription on the three test clips — before any translation.

SHORT

“What's a lesson you wish you learned earlier in life?”

LONG

“Raise your hand if you have recently had to introduce yourself or make small talk. Yes, that is what today is all about. How to speak better in the moment.”

NOISY

“It it's all connected if we go back and look at it and there's a whole lot of... I thought that was the end. Well, it was the end, but it was the beginning of this thing...”

Results — Arabic

English transcript translated to Arabic, then spoken via gTTS.

SHORT

ما هو الدرس الذي تتمنى أن تتعلمه في وقت سابق من حياتك؟

LONG

ارفع يدك إذا اضطررت مؤخرًا إلى تقديم نفسك أو إجراء محادثة قصيرة. نعم، هذا هو ما يدور حوله اليوم. كيف تتحدث بشكل أفضل في هذه اللحظة

NOISY

كل شيء متصل إذا عدنا ونظرنا إليه وهناك الكثير من الأشياء التي اعتقدت أنها كانت النهاية... لكنها كانت بداية هذا... الشيء

Results — French

English transcript translated to French, then spoken via gTTS.

SHORT

Quelle est la leçon que vous auriez aimé apprendre plus tôt dans la vie ?

LONG

Levez la main si vous avez récemment dû vous présenter ou bavarder. Oui, c'est de cela qu'il s'agit aujourd'hui. Comment mieux parler sur le moment.

NOISY

Tout est lié si nous y retournons et y regardons... Je pensais que c'était la fin. Eh bien, c'était la fin, mais c'était le début de cette chose...

Results — Spanish

English transcript translated to Spanish, then spoken via gTTS.

SHORT

¿Cuál es una lección que desearías haber aprendido antes en la vida?

LONG

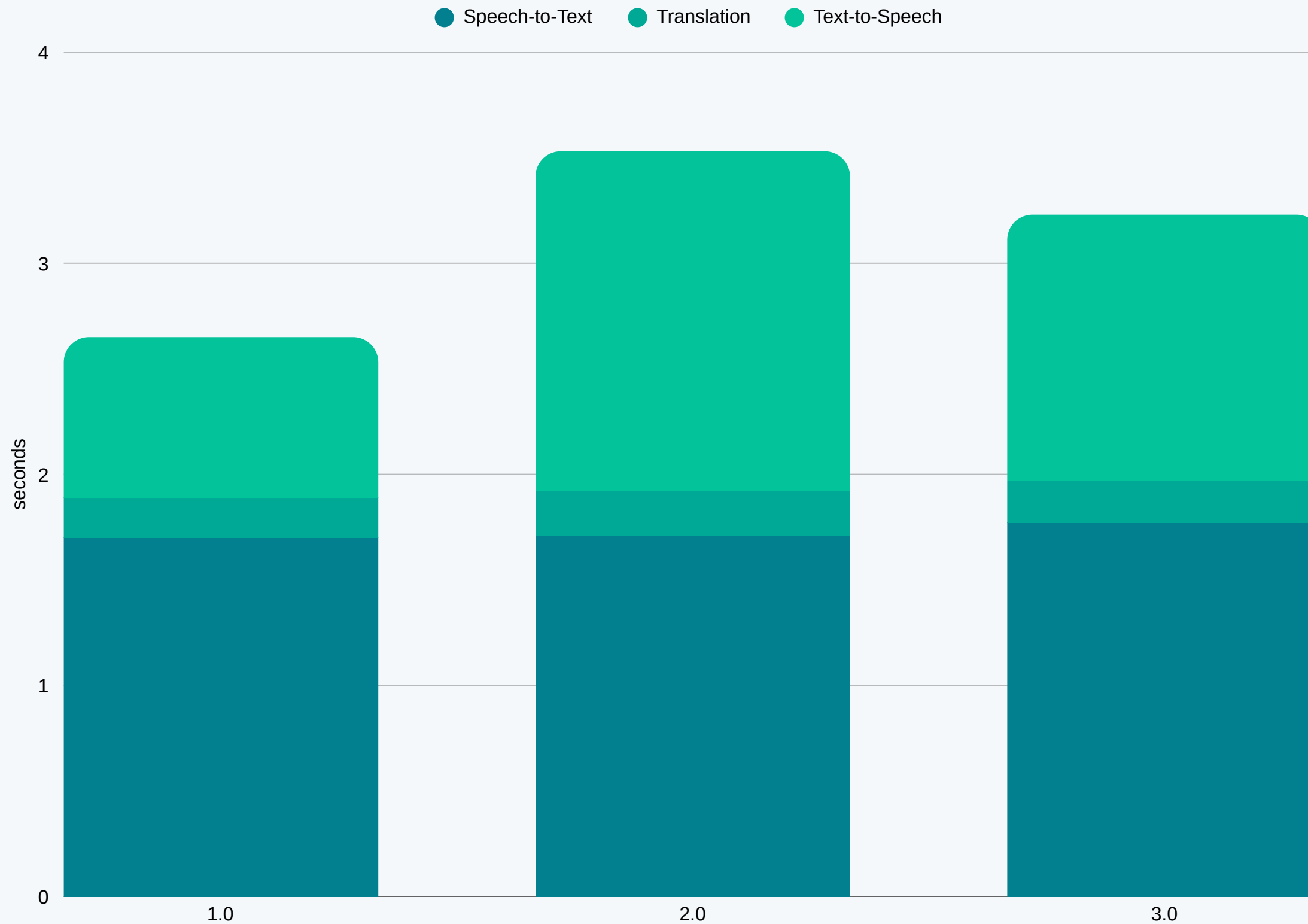
Levante la mano si recientemente ha tenido que presentarse o entablar una pequeña charla. Sí, de eso se trata hoy. Cómo hablar mejor en el momento.

NOISY

Está todo conectado si retrocedemos y lo miramos y hay muchas cosas que pensé que era el final. Bueno, fue el final, pero fue el comienzo de esto...

System Latency

End-to-end timing on the noisy clip, measured per stage for each target language.



2.7–3.5s

total pipeline per clip



~1.7s

Speech-to-Text (stable across languages)



0.2–0.25

real-time factor — ~4–5× faster than the audio

Real-Time, Long Audio & Context



Is it real-time capable?

No — the pipeline is offline / batch, not streaming. Each stage needs its predecessor's complete output (full audio → full transcript → full text). Computation is fast, but the architecture is built for file-based translation, not live input.



How does it handle long audio?

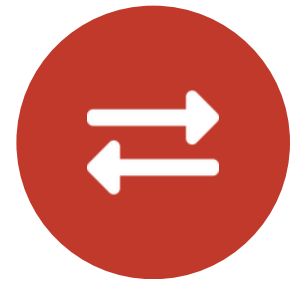
Well. Whisper splits long audio internally into 30-second windows, decodes each, and stitches them together — so length isn't a problem. The long clip transcribed cleanly end-to-end.



How does it handle context-dependent words?

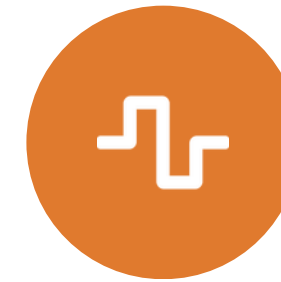
Every stage is context-aware. Whisper disambiguates homophones (their/there) from surrounding words; Google Translate uses sentence context for gender, agreement & idioms. Tradeoff: errors compound — a mis-transcribed word is faithfully translated and spoken.

Challenges & Limitations



Errors compound across stages

There's no cross-stage correction. A word Whisper mis-hears is faithfully translated and spoken — the mistake propagates to the final audio.



Noise sensitivity

Accuracy drops on noisy input. Disfluencies and repeated words (“It it's”) pass straight through, since the system has no dedicated noise-handling step.



Not real-time / streaming

The pipeline is offline and batch. Each stage must fully finish before the next begins, so it can't process speech live as someone is talking.



Dependence on external services

Translation and TTS call Google services, so output quality, available languages, and voices are bound to those APIs and require connectivity.



Future Work — Going Real-Time

How to evolve the offline pipeline into a live, streaming translator.



Core shift: stop waiting for whole files — process speech in small, overlapping chunks as it arrives, so all three stages run concurrently instead of in sequence.



01 Streaming audio capture

Get audio from microphone real time instead of loading a finished file.



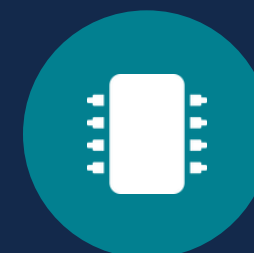
02 Incremental / streaming ASR

Use a streaming-capable model (e.g. faster-whisper, whisper-streaming, or wav2vec2) that emits partial transcripts as you speak.



03 Pipeline the stages

Run STT, translation and TTS in parallel on a queue so a chunk is translated while the next is still being transcribed.



04 Low-latency TTS + GPU

Swap gTTS for a local streaming TTS (e.g. Coqui/Piper) and keep models GPU-resident to cut network and synthesis delay.



Thank You

Speech-to-Text · Translation · Text-to-Speech

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